experiment and I repeated it a few more times, with the same result. Now, Kalam wanted to see up close what was happening to the gunpowder in the jar. The jar was kept on a table in the middle of a room, with the vacuum pump at a distance. Kalam went close and peered into the jar, his nose pressed against the glass, and asked me to fire it. If the gunpowder fired, it would smash the jar to smithereens; but we knew by now it wouldn't. And Kalam wanted to witness the inertness of gunpowder with naked eyes. The countdown started: 5, 4, 3 … And then, I noticed that the vacuum pump was not properly connected to the jar. This meant that the pressure would not be low enough for the gunpowder to fire. In other words, it would fire in Kalam's face.

'Stop, don't ignite the squib,' I shouted, and without waiting for the assistant to hear me, leapt over Kalam, bringing him and myself on to the floor. In another second, there was a blast, glass shreds flew all over, I was on top of Kalam.

After the smoke had settled, Kalam sat up and smiled: 'See, it fired.'

This man's enthusiasm for experiments was so much that he was unaware of the potential disaster he had just escaped. I replied, 'See, the vacuum pump is not connected.' I joked that we now not only know that gunpowder does not fire at low pressure, but also that it does fire at normal pressure. The smile on our faces soon vanished when we realised that the sounding rocket with G20 gunpowder could not be a success. Now, who would break the bad news to director Murthy?

There wasn't much choice, and Kalam and I went to the director and explained why the launch could not be a success. Murthy asked me why I did not consider this possibility earlier while designing the igniter. My answer was simple: I did not know about the behaviour of gunpowder at low pressure till the dusty file containing the scientific paper of Able and Noble fell on my lap.

There was no way we could replace the gunpowder with something else at such short notice, and the ground stations across the world were waiting for the launch. Do we call off the launch? That was the big question Murthy had to answer. Calling off the launch would mean that the world would